

$$P = P_{n-1} \dots P_1$$

P_i is orthogonal matrix.

$$PA = LU$$

$$MA = U \quad M = (M_{n-1}P_{n-1} \dots M_1P_1)$$

$$\downarrow$$

$$A = M^{-1}U.$$

$$PM^{-1}U = LU$$

$$L = PM^{-1} = P(P_1^{-1}M_1^{-1} \dots P_{n-1}^{-1}M_{n-1}^{-1}) =$$

$$= (P_{n-1} \dots P_1 P_1^{-1}M_1^{-1} \dots P_{n-1}^{-1}M_{n-1}^{-1}) = P_{n-1} \dots P_2 M_1^{-1} P_2^{-1} M_2^{-1} \dots P_{n-1}^{-1} M_{n-1}^{-1}$$

$$= P_{n-1} \dots (P_2 M_1^{-1} P_2^{-1}) M_2^{-1} \dots P_{n-1}^{-1} M_{n-1}^{-1}$$

P_i - changes j th row ^{with} the row corresponding to highest pivot row _{$i > j$} when multiplied from left. When ^{P_i^{-1}} multiplied from right it changes i th column with j th. proof:

$$AP_i^{-1} = AP_i^T = (P_i A^T)^T$$

~~P_i^2 proof: changing i th and j th row twice means not changing the matrix.~~

$P_i A^T$ changes rows of $A^T \rightarrow$ changes columns of A .

$$M_k = I - m_k e_k^T$$

$$P_i^2 A = A$$

$$m_k = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ m_{k,k+1} \\ \vdots \\ m_{k,n} \end{bmatrix}_{k+1} \quad e_k = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}_k$$

$$M_k^{-1} = I + m_k e_k^T$$

$$(I + m_k e_k^T)(I - m_k e_k^T) = I + \overbrace{m_k e_k^T - e_k^T m_k}^0 - (e_k^T m_k)(m_k e_k^T) = I$$

$$e_k^T m_k = 0$$

$$P_2 M_1^{-1} P_2^{-1} = P_2 (I + m_1 e_1^T) P_2^{-1} = I + P_2 m_1 e_1^T P_2^{-1} = I + P_2 m_1 e_1^T$$

$$e_1^T P_2^{-1} = (P_2 e_1)^T = e_1^T$$

$$P_2 \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \Rightarrow e_1 \text{ no matter what } i \text{ is } \neq i, \text{ so changing } i \text{ with 2th element}$$

$$P_{n-1} \dots P_3 (I + P_2 m_1 e_1^T) M_2^{-1} P_3^{-1} \dots P_{n-1}^{-1} M_{n-1}^{-1} \quad 0 \text{ means no change.}$$

$$(I + P_2 m_1 e_1^T) M_2^{-1} = (I + P_2 m_1 e_1^T) (I + m_2 e_2^T) =$$

$$= I + P_2 m_1 e_1^T + m_2 e_2^T + P_2 m_1 (e_1^T m_2) e_2^T$$

$$= I + P_2 m_1 e_1^T + m_2 e_2^T \quad e_1^T m_2 = 0$$

$$= I + m_1' e_1^T + m_2 e_2^T$$

$$= (I + m_1' e_1^T) (I + m_2 e_2^T) =$$

$$= M_1^{-1} M_2^{-1}$$

Lower triangular.

$$[1 \ 0 \ 0 \ \dots \ 0] \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = m_1'$$

this will change just the 2nd element with same $j \rightarrow 2\text{th element.}$

Inductivity, we proved that.

if we have ~~lower triangular~~ M_i then

$$P_{i+1} M_i^{-1} P_{i+1}^{-1} = P_{i+1} (I + P_i \dots P_2 m_1 e_1^T + P_i \dots P_3 m_2 e_2^T + \dots + m_i e_i^T) P_{i+1}^{-1}$$

if we have matrix of form

$$X_i = I + P_i \dots P_2 m_1 e_1^T + P_i \dots P_3 m_2 e_2^T + \dots + m_i e_i^T$$

$$\text{then } X_{i+1} = I + P_{i+1} \dots P_2 m_1 e_1^T + \dots + m_{i+1} e_{i+1}^T$$

$$\text{where } X_{i+1} = P_{i+1} X_i M_i^{-1} P_{i+1}^{-1} = P_{i+1} X_i (I + m_i e_i^T) P_{i+1}^{-1}$$

$$P_{i+1} X_i P_{i+1}^{-1} + P_{i+1} (X_i m_i e_i^T) P_{i+1}^{-1} = P_{i+1} X_i P_{i+1}^{-1} + P_{i+1} m_i e_i^T$$

~~X_i~~

$$X_i (m_i e_i^T) = X_i (m_i e_i^T) = m_i e_i^T$$

$$e_j^T m_{i+1} = 0 \text{ for all } j < i \quad e_j^T m_i = 0 \text{ for all } j < i, \text{ so}$$

$$X_{i+1} = P_{i+1} X_i P_{i+1}^{-1} M_{i+1}$$

$$X_{i+1} = P_{i+1} X_i P_{i+1}^{-1} M_{i+1} = P_{i+1} X_i P_{i+1}^{-1} (I + m_i e_i^T) =$$

$$= (I + P_{i+1} \dots P_2 m_1 e_1^T + \dots + P_{i+1} m_i e_i^T) (I + m_{i+1} e_{i+1}^T) =$$

multiplying e_s with P_k from right doesn't change

e_s for $k > s$, so, we will ~~omit~~ consider.

$$e_s^T P_{i+1} = e_s^T$$

$$= (I + P_{i+1} \dots P_2 m_1 e_1^T + \dots + P_{i+1} m_i e_i^T) + (m_{i+1} e_{i+1}^T) = X_{i+1}$$

other ~~term~~ terms go to zero because.

$$m_a e_a^T (m_b e_b^T) = (e_a^T m_b) m_a e_b^T$$

$$e_a^T m_b = 0$$

for $b > a$,

because $i+1 > j \in (1, i)$

we omit those term

because

let's define.

$$X_2 = I + m_1 e_1^T + m_2 e_2^T$$

so, getting back to our question,

$$L = P M^{-1} = X_{n-1} = I + P_{n-1} \dots$$

$$= I + P_{n-1} \dots P_2 m_1 e_1^T + \dots P_{n-1} m_{n-2} e_{n-2}^T + m_{n-1} e_{n-1}^T$$

multiplying m_i with " P_k " sequentially where

$$k > i, \quad P_{n-1} \dots P_{i+1} m_i = m_i'$$

means just

permutating

elements s where $i < s < n$, so, m_i' is still of the form

so, we can say,

$$I + m_1' e_1^T + \dots + m_i' e_i^T + \dots + m_{n-1}' e_{n-1}^T =$$

$$= (I + m_1' e_1^T) (I + m_2' e_2^T) \dots (I + m_{n-1}' e_{n-1}^T)$$

$$= \boxed{M_1^{-1} M_2^{-1} \dots M_{n-1}^{-1}} \rightarrow \text{which is of lower form}$$

Because.

L^{-1} is also lower.

$$A \quad I$$

$$\begin{bmatrix} a_{11} & 0 & 0 & \dots & 0 \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \dots & a_{nn} & 0 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix}$$

we can see that when we apply elementary

matrices $E_k \dots E_1$ to A to I

where $E_k \dots E_1$ reduces A to U ,
we only change lower triangle part of

I

